Guide: Using Libra Cluster with VS Code (Python & Jupyter on GPU)

This guide explains how to run Python scripts and Jupyter notebooks on the **Libra Cluster** using GPU resources, with **VS Code Remote–SSH**. You already have remote SSH set up, so this document focuses on requesting GPU nodes, running Python files, and using Jupyter Notebooks.

1. Connect to Libra with VS Code

1. Open VS Code.
2. Use the Remote–SSH extension to connect to: your_username@libra.slu.edu.
3. You will land on the login node (not suitable for GPU work).

2. Running Python Scripts on a GPU Node

1. From VS Code's integrated terminal, request an interactive GPU allocation:
2. `srun --partition=gpu --gres=gpu:1 --time=02:00:00 --cpus-per-task=4 --mem=16G --pty bash`
3. Load required modules (adjust versions as needed): `module load cuda/11.7` and `module load python/3.10`.
4. Activate your conda/venv environment: `conda activate mygpuenv`.
5. Run your script with GPU support: `python myscript.py`.

3. Running Jupyter Notebooks on a GPU Node

1. Request a GPU node (same `srun` command as above).
2. Activate your environment with GPU-enabled Python libraries.
3. Launch Jupyter without a browser: `jupyter notebook --no-browser --port=8888 --ip=0.0.0.0`.
4. Copy the URL with the token (e.g. `http://gpu-node123:8888/?token=abcd...`).
5. In VS Code, open your `.ipynb` file. Click **Select Kernel** → **Existing Jupyter Server** → paste the URL.
6. Your Jupyter notebook in VS Code now runs on the GPU node.

4. Verifying GPU Access

Inside your Python script or Jupyter notebook, run:

```
```python import torch print(torch.cuda.is_available())
print(torch.cuda.get_device_name(0)) ```
```

5. Notes & Best Practices
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- Always request resources with `srun` or `sbatch`. Never run heavy workloads on the login node.
- Use the correct partition name (`gpu`) as described in Libra’s documentation.
- Check job limits (max time, memory, GPUs) in the official Libra user guide.
- Use `nvidia-smi` to check GPU usage inside a session.
- For long jobs, prepare a batch script (`.sbatch`) instead of interactive `srun`.